Lab Report – EE-170L Computer Programming (Lab 03)

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LAB 3	(programming)
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1. Objective

The purpose of this lab is to understand and implement:

- **Comparison/Relational Operators**
- **Logical Operators**
- **Decision-making structures (if, if-else, else-if)**

2. Theory

2.1 Relational Operators

Relational operators compare two values and return a boolean result (true or false).
Examples:

- < : Less than
- > : Greater than
- <= : Less than or equal to
- >= : Greater than or equal to
- == : Equal to
- != : Not equal to

2.2 Logical Operators

Logical operators combine or negate conditions.

- ! : NOT
- && : AND
- || : OR

2.3 Decision Control Structures

- **if statement**: Executes a block if condition is true.
- **if-else statement**: Executes one block if condition is true, otherwise another block.
- **else-if statement**: Allows multiple conditions to be checked sequentially.

3. Programs and Results

Task 1: Comparison Operators

```
#include <iostream>
using namespace std;

int main () {
   int a, b;
```

```
    cout << "Enter two numbers: ";
    cin >> a >> b;

    if (a == b)
        cout << "Both numbers are equal." << endl;
    else if (a > b)
        cout << "First number is greater." << endl;
    else
        cout << "First number is smaller." << endl;

    return 0;
}
```

Sample Output:

```
Enter two numbers: 10 20
First number is smaller.
```

Task 2: Logical Operators

```
#include <iostream>
using namespace std;
```

```
int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;

    if (num > 0 && num < 100)
        cout << "Number is between 0 and 100." << endl;
    else
        cout << "Number does not meet the condition." << endl;

    return 0;
}
```

Sample Output:

```
Enter a number: 50
Number is between 0 and 100.
```

Task 3: If Statement

```
#include <iostream>
using namespace std;
```

```
int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;

    if (num > 0)
        cout << "Number is positive." << endl;

    return 0;
}
```

Sample Output:

```
Enter a number: -5
(no output since condition is false)
```

Task 4: If-Else Statement

```
#include <iostream>
using namespace std;

int main() {
    int age;
```

```
    cout << "Enter your age: ";
    cin >> age;

    if (age >= 18)
        cout << "You are eligible to vote." << endl;
    else
        cout << "You are not eligible to vote." << endl;

    return 0;
}
```

Sample Output:

```
Enter your age: 16
You are not eligible to vote.
```

Task 5: Else-If Statement

```
#include <iostream>
using namespace std;

int main() {
    int num;
```

```
    cout << "Enter a number: ";
    cin >> num;

    if (num > 0)
        cout << "Number is positive." << endl;
    else if (num < 0)
        cout << "Number is negative." << endl;
    else
        cout << "Number is zero." << endl;

    return 0;
}
```

Sample Output:

```
Enter a number: 0
Number is zero.
```

4. Discussion

- Relational operators allow comparison between values, essential for decision-making.
- Logical operators combine conditions, making programs more flexible.

- Control structures (if, if-else, else-if) enable branching logic, allowing programs to respond differently based on input.

5. Conclusion

In this lab, I successfully implemented relational operators, logical operators, and conditional statements in C++. These concepts are fundamental for writing programs that make decisions based on user input.